

Program Name: MCA	Semester: IV
Subject Name: Block Chain	Subject Code:
End Sem Exam (MM): 30	Internal Assessment (MM): 70
Credits: 4	Contact Hours: 60

Course Objectives: Understand blockchain fundamentals, cryptographic principles, Ethereum and Bitcoin technologies, and applications in diverse domains like healthcare.

Detailed Syllabus:

S. No.	UNITWISE DETAILS	COURSE OUTCOME	BLOOM'S LEVEL	RELEVANCE NEED	CONTACT HOURS
1	UNIT-1: Introduction to Blockchain: Need for Distributed Record Keeping, Blockchain architecture, blockheader detailed design, Abstract Models for Blockchain, Proof of Work (PoW), liveness and fairness, Proof of Stake (PoS) based Chains, Hybrid models (PoW + PoS); Types of Blockchain..	CO1: Understand blockchain fundamentals, including architecture, PoW, PoS, and models	Understanding	Global	9
2	UNIT-2: Blockchain Consensus Algorithm challenges and solutions, Modeling faults and adversaries, Byzantine Models of Fault tolerance; Zero Knowledge proofs and protocols in Blockchain.	CO2: Understand consensus algorithms, fault modeling, Byzantine models, and zero-knowledge proofs	Understanding	Global	9
3	UNIT-3: Introduction to cryptographic basics for cryptocurrency - a short description of Hashing, digital signature schemes, encryption schemes and elliptic curve cryptography, verifiable random functions.	CO3: Understand cryptographic basics: hashing, signatures, encryption, elliptic curves, randomness	Understanding	Global	9
4	UNIT-4: Blockchain 2.0: Introduction to Ethereum, Ethereum Virtual Machine (EVM), Wallets for Ethereum, Solidity, Smart Contracts, Attacks on smart contracts, The Turing Completeness of Smart Contract Languages and verification challenges. Blockchain 3.0: Hyperledger implementation on	CO4: Understand Ethereum, smart contracts, and Hyperledger for blockchain solutions	Applying	Global	9

	Ethereum, the plug and play platform and mechanisms in permissioned blockchain.				
5	UNIT-5 Application of Blockchain: Bitcoin- Bitcoin consensus, Wallet, Bitcoin Blocks, Merkley Tree, hardness of mining, transaction verifiability, anonymity, forks, double spending, mathematical analysis of properties of Bitcoin. Altcoins. Medical record management systems, DNS records.	CO5: Analyze Bitcoin's blockchain applications, consensus, and related concepts	Analyzing	Global	9
Total Hours				45	

Text books:

- Andreas M. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies, O'Reilly Media, 2014.
- Daniel Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Reference books:

- Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System, self-published, 2008.
- Chris Dannen, Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginners, Apress, 2017.
- David C. Lee, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2017.
- Pradeep Gohil, Blockchain for Business: A Practical Guide, Springer, 2021.